

9708/31/o/n/20

1-D:

The correct answer is D: **to find the highest positive difference between total benefit and total cost.**

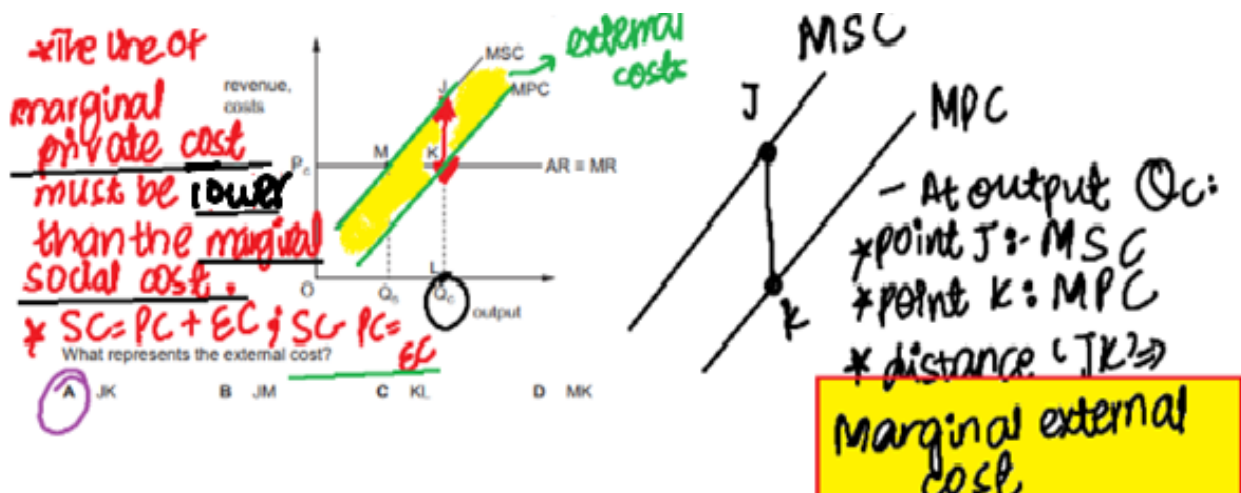
The main aim of government cost-benefit analysis is to evaluate the desirability of a project or policy by comparing the total benefits (**social benefits**) to the total costs (**social costs**). The goal is to determine whether the benefits outweigh the costs, and to identify the option with the highest net social benefit.

A: While firms may aim to maximize private benefits, they don't consider external costs into consideration.

B: Equating total social benefit to total social costs is not the objective of cost benefit analysis; instead, it aims to find the project with highest positive difference between social costs and social benefits.

C: Internalizing external costs through taxation represents government intervention to solve the problem of market failure associated with over-allocation of demerit goods.

2-A:

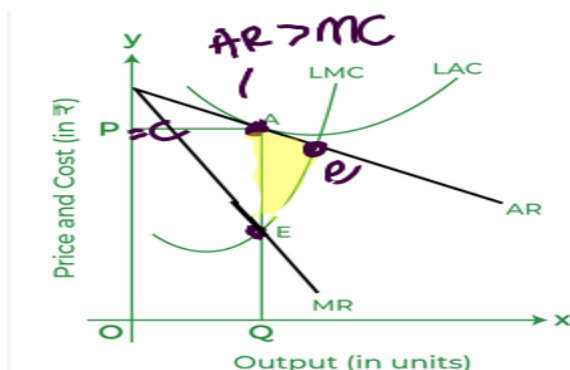


3-A:

In the long-run, a monopolistically competitive firm earns normal profits or zero-economic profits because at the profit maximizing equilibrium (where $MC=MR$), Average revenue is equal to Average total cost. **This means that supernormal profit is not earned in the long-run.**

Moreover, since it produces where $MC=MR$ (profit maximizing level of output), it can be said that it produces less than the allocatively efficient level of output (which is determined through the equilibrium where $MC=AR$). As a result of which, it can be said that it is allocatively inefficient.

This can be seen in the diagram below.



4-B:

biscuits / chocolates eaten	total utility (units)		
	biscuits	chocolates	
1	20	10	
2	38	20	
3	52	32	
4	62	38	
5	70	42	
6	74	42	
7	74	40	

A – After eating the first chocolate, the marginal utility remains constant at “10” when two chocolates are consumed, and then rises to “12” chocolates, upon the consumption of the third chocolate.

B – Yes, this is correct as the consumer gets **negative marginal utility of -2** from the seventh chocolate eaten.

C – This is impossible that a consumer chooses **to never eat chocolates when biscuits are available**. This is because, more consumption of biscuits will lead to the marginal utility to diminish, which will cause the rational consumer to switch to chocolates.

D – The second biscuit gives the **marginal utility of “18”**, which is not the same as the fourth chocolate, which gives the marginal utility of “6”

5-B:

- ➔ Because of a rise in income, the quantity of Y demanded has contracted, which represents a negative relationship between income and demand, and this means that **good Y is an “inferior good”**
- ➔ Because of a rise in income, the quantity demanded of good X has risen, which represents the case of **“normal goods”** whereby there is a positive relationship between income and demand.

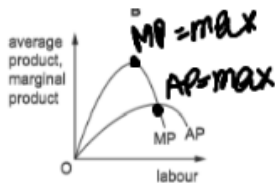
- ★ $FC = 1000$ Yuan
- ★ $AVC = 20$ Yuan
- ★ $ATC = 25$ Yuan

$$\begin{aligned} \text{ATC} &= \text{AFC} + \text{AVC} \\ 15 &= \frac{\text{TFC}}{Q} + 10 \\ 15 &= \frac{1000}{Q} + 10 \\ 15 - 10 &= \frac{1000}{Q} \\ 5 &= \frac{1000}{Q} \Rightarrow Q = \frac{1000}{5} \Rightarrow Q = 200 \end{aligned}$$

7- D:

A: Yes, it is a requirement for at-least one of the factors of production to be fixed, and that is mainly capital.

B:Yes, **average returns to a variable factor** or “**Average product of labor**” will increase first then fall(**when the Marginal product of labor is lesser than the average product of labor**). This can be seen in the diagram below, and also in the table below.



It is the "marginal product of labour" which reaches its maximum the first.

Regardless of whether marginal itself is rising or falling, the average product of labour continues to rise or slope upwards, as long as the marginal product of labour is above it, or greater than it.

Similarly, the average product of labour falls as soon as the marginal product of labour is lesser than the average product of labour, this is why, the intersecting point of marginal and average represents the "maximum of average"

Therefore, the API is the second to reach the maximum.

C :Similarly, the marginal product of labor will also rise first then fall, since the law of diminishing returns sets in, this can be seen in the diagram above, and the table below.

	Number of labour						
	0	1	2	3	4	5	6
TP	0	50	120	180	200	200	180
APL	0	50	60	60	50	40	30
MPL	0	50	70	60	20	0	-20

D: Inputs of all factors can only be variable in the long-run, and in the short-run, at least one factor is assumed to be fixed, which is mainly capital. This is why, key D, becomes correct.

8- C:

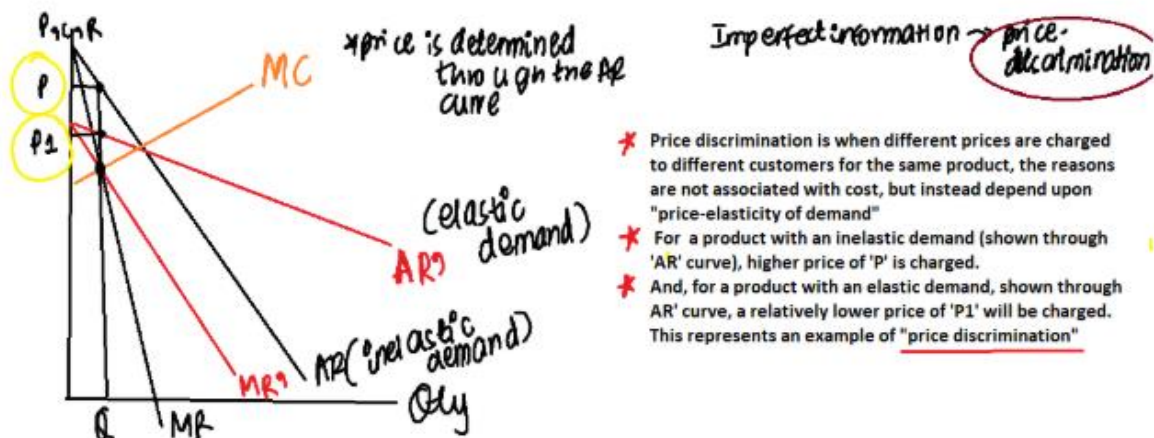
A – The firm is shown to make supernormal profits because Average revenue is greater than Average cost. Therefore, it is incorrect to say that it is selling below cost to keep out potential entrants.

B – If the firm wished to maximize the volume of sales , or the units sold, this means that it will charge the lowest price it possibly can. This means that it will operate where normal profits are made, that is when, the Average revenue is equal to Average cost.

C – At output of Q_e , since marginal revenue is zero, this means that the total revenue is maximized. Managers aim for “sales revenue maximization” because managerial salaries and bonuses are linked to sales revenues rather than profits, which in turn gives existence to the “principal agent problem”. This is why, key C , is correct.

D – Managers wanting to have a large team working for them to increase their status does not explain the objective of “sales revenue maximization” either.

9-B:



A – Even if there are many buyers, price discrimination cannot be done if there is availability of information.

B – It can be seen in the diagram above that, there must be markets with “different price-elasticities of demand”, in order for price discrimination to be done.

C – Price discrimination is mainly “done for the same product”.

D – Even if markets are separate, price discrimination cannot be done, until and unless there are differences in price-elasticities of demand in each market. Additionally, for price discrimination to be effective, there must be “imperfect information”

10-B:

A – The law of diminishing marginal utility occurs when the marginal utility, that is the satisfaction derived upon consuming an additional unit, declines or diminishes.

B _ As more units of a variable input are added in the short run, initially, the total product (output) increases at an increasing rate. This is reflected by a steeply rising portion of the total product curve. **However, there comes a point where adding more of the variable input leads to smaller increases in output. At this stage, the marginal product starts to decline, indicating the onset of diminishing returns. The short-run average cost curve, which is derived from the total cost and total output, typically exhibits a U-shape. The upward sloping portion of the average cost curve reflects increasing average costs due to diminishing returns.** This occurs because additional units of the variable input lead to smaller increases in output, causing the **average cost per unit to rise.**

C- This is incorrect as fixed costs remain unchanged in the “short-run”

D – Technical dis-economies of scale occur in the “long-run”

11-C:

In case of predatory or destroyer pricing, the firm charges very low prices in the short-run, or in other words, prices fall in the short-run. And, after competition is eliminated, prices are raised in the long-run. This is why, key C, becomes correct.

12-B:

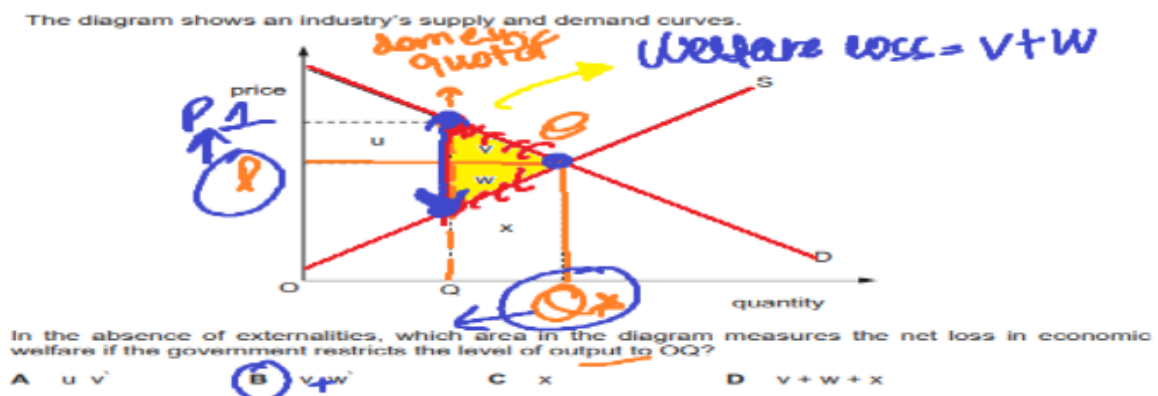
A: Small firms continue to exist in a market dominated by large firms since they provide personalized, specialist services due to their smaller-size, which large firms cannot provide.

B: This is not a reason behind the existence of small firms. Instead, large firms exist because of “economies of scale”

C: Innovation can provide small firms with a competitive advantage over their competitors. By developing new products, services, or processes, these firms can differentiate themselves in the market dominated by large firms.

D: Good knowledge of the local markets can help them meet the needs of the consumers more successfully, leading to them being able to exist along-side large firms.

13- B



14 – C:

A- As a result of government's attempts to ban cigarettes in enclosed public spaces such as: offices, shops and bars , if there is a decreased consumption of cigarettes, this represents the "success of government intervention"

B- If this results in a decreased level of passive smoking by non-smokers as a results of less smoking being done , this will lead to a reduction in external costs, which again represents the "success of government intervention"

C- If cigarettes are banned inside enclosed public spaces such as: offices and bars, they will be consumed outside, which will subsequently result in an increased litter from used cigarettes outside offices and shops. This is correct because it results in "government failure".

D- Again, when due to a decrease in consumption of cigarettes, when individuals switch to less dangerous substitutes like: nicotine gums and patches, this represents the "success of the government intervention"

15- A:

A – When a country's tax system is made more progressive, this means that a greater proportion of the incomes of the higher income groups will be taken in tax, as a result of which, there will be funds available with the government for redistribution, which will subsequently improve equality.

B – When someone is made better off without making someone else worse off, this represents "Pareto improvement"

C – When the best use is made of scarce resources, this represents efficiency.

D – An increase in gini-coefficient represents inequality.

16- C:

A: A decrease in income tax may boost aggregate demand, but it doesn't improve occupational mobility. While increased Aggregate demand may lead to more job opportunities, it doesn't enable workers to transition between occupations.

B: Decreasing unemployment benefits may incentivize job searching thereby reducing "voluntary unemployment", but it doesn't enhance occupational mobility.

C: An increase in retraining schemes for the unemployed can improve occupational mobility by enhancing skills and qualifications. This in turn enables workers to transition more easily between occupations, and since now they can easily and flexibly switch in between occupations when the structure of the economy is changing, this will reduce structural unemployment.

D: More advertising of employment vacancies can increase job awareness thereby reducing search unemployment which is categorized under frictional un-employment, but it doesn't address the skills or qualifications needed for occupational mobility. While workers may become aware of more job openings, they may still lack the necessary skills to transition to a new occupation.

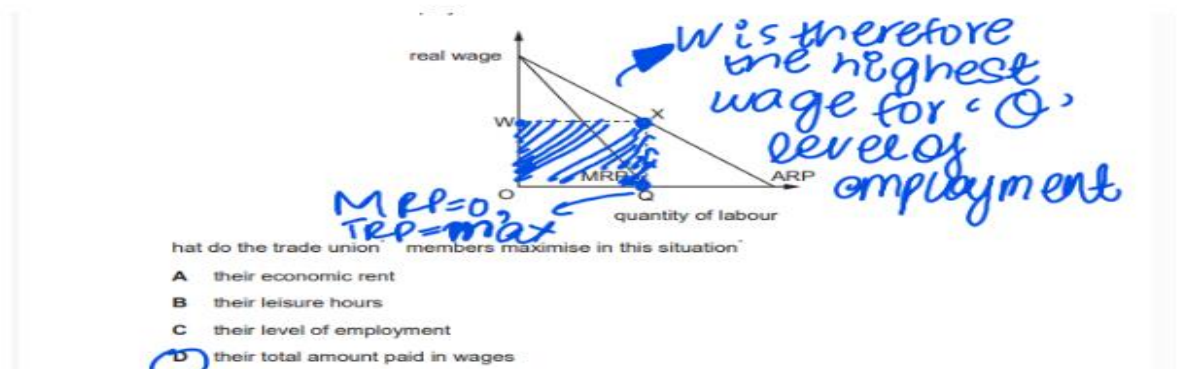
17-D:

A – Trade union members do not maximize their economic rent because economic rent is shown through the region above the supply curve and below the wage rate. It is maximized when the supply of labor is perfectly inelastic.

B – We can not determine if leisure hours are maximized or not, because the information about total hours and working hours is not available.

C – This is incorrect because quantity of labor can still increase.

D – At “Q” level of employment, Marginal revenue product is equal to zero, which means that the total revenue product will be at its maximum, representing that the wage rate derived from the ARP curve will be the highest for the quantity “Q”. This can be seen in the diagram below. **Therefore, the total amount paid in wages will be the maximum.**



This is why, key D, becomes correct.

18 – A:

Human Development Index includes:

GNI per capita

Health care: Life expectancy at birth

Education: Mean years of schooling and expected years of schooling

Since infant mortality rate(A) is not included, option A becomes correct.

19- C:

If a developing country permits repatriation of profits, embarks on a programme of privatization and creates genuine independence for their central banks, this will attract foreign direct investment in form of multi-nationals in manufacturing.

However, if the country lacks “well-developed infrastructure”, it will not attract fdi in manufacturing even if there are low-wages and low-cost.

This explains why key C is correct.

20- B:

B - Frictional unemployment refers to temporary unemployment due to the time it takes to match workers with job openings. This type of unemployment is a normal part of the job search process and is usually short-term. In this case, the individual found a job quickly, indicating that their unemployment was brief and likely due to the ordinary search process of finding a new job.

A: Cyclical unemployment is related to economic downturns or recessions, which is not indicated in this scenario.

C: Seasonal unemployment refers to job losses due to seasonal fluctuations in industries like agriculture or tourism, which doesn't apply here.

D: Structural unemployment occurs when there is a structural change in the economy, mainly in the pattern of the demand supply of the economy, especially when workers are geographically and occupationally immobile.

21- D:

A – A low level of GDP did not mean inability to build stocks of wealth. An example of this is Kenya, which although had the lowest level of gdp per head as compared to other countries, still managed to build second highest stocks of wealth in terms of natural, human and produced resources.

B – All countries except for Kenya faced depletion of natural resources.

C – This is incorrect because although Russia did not experience a growth in IWI, it still had the second highest GDP per head.

D – This is correct because all countries faced an increase in human resources.

22- D:.

D - Monetarists believe that inflation is primarily caused by excessive money supply growth. To combat inflation, monetarists advocate for reducing the money supply.

The sale of securities on the open market:-

- Reduces money supply (the amount of money in circulation) -> Increases interest rates-> Reduces consumption and investment -> Decreases aggregate demand -> Curbs inflationary pressures

A: Increasing indirect taxation could lead to higher prices of goods and services, exacerbating inflation. Additionally, it is a part of fiscal policy which involves the government manipulating taxes and government spending to influence Aggregate demand.

B: Monetary policy involves the government manipulating money supply, interest rates and exchange rate to influence Aggregate demand. Direct foreign exchange rate intervention does not represent contractionary monetary policy unless revaluation is done.

C: Introducing maximum prices is a price control measure, which monetarists generally oppose as they believe it distorts market mechanisms.

23- D:

A consumption function tilts when there is a change in the slope of the function, representing a change in the marginal propensity to consume (MPC). This means that the relationship between income and consumption has changed, but the autonomous consumption (the intercept) remains the same. **On the other hand, a consumption function shifts when there is a change in autonomous consumption, represented by a shift of the entire function upward or downward, while the slope (MPC) remains constant.**

- Tilt: If the consumption function was originally $C = 500 + 0.8Y$ and the MPC increases to 0.9, the new function would be $C = 500 + 0.9Y$. The function has tilted, with a steeper slope.
- Shift: If the consumption function was originally $C = 500 + 0.8Y$ and autonomous consumption increases to 600, the new function would be $C = 600 + 0.8Y$. The function has shifted upward, with the same slope.

A - A minimum deposit requirement for credit purchases would reduce consumption, as consumers would need to pay a portion of the purchase price upfront. This would decrease consumption and lead to a downward shift of the consumption function, opposite of the desired shift from C1 to C2.

B – An increase in direct taxation represents an increase in marginal propensity to tax (mpt), as a result of which, the marginal propensity to consume falls, which represents that the consumption function tilts downwards.

C - A higher interest rate would increase the cost of borrowing, making consumers less likely to take out loans for consumption purposes. This would lead to a reduction in consumption, causing a downward shift of the consumption function, opposite of the desired shift.

D - Expecting an increase in indirect taxes (such as sales tax or value-added tax), consumers may accelerate their purchases to avoid paying higher prices in the future. This would lead to an increase in autonomous consumption, shifting the consumption function upward from C1 to C2. This explains why key D is correct.

24- D:

According to the accelerator principle, investment is a function of rate of change of income. The formula to find out the coefficient of accelerator is :- Change in Investment/ Change in Income. This means that if income is increasing at a rapid rate (i.e. ΔY is large), investment will be high, as businesses will need to expand capacity to meet growing demand. Conversely, if income growth slows down (i.e. ΔY is small), investment will be lower, as businesses may not need to invest as much in new capacity.

This explains why key D is correct.

25 – C:

Private sector ($S > I$): Yes

In order for there to be net leakages in the circular flow, saving (leakage) should be greater than investment (injection)

Govt sector ($T > G$): Yes

In order for there to be net leakages in the circular flow, tax (leakage) should be greater than government spending (injection)

Trade sector ($X > M$): No

In order for there to be net leakages in the circular flow, export revenues that represent an injection into the circular flow should be lesser than import expenditures that represent a leakage from the circular flow.

26- D:

The Keynesian monetary transmission mechanism suggests that changes in the money supply affect the economy through interest rates, which in turn influence investment and aggregate demand.

1. **Sell securities** → **Reduce money supply** (central bank sells securities, reducing the money supply)
2. **Reduce money supply** → **Increase interest rates** (less money supply relative to unchanged money demand leads to a shortage of money in the money market, as a result of which, there is an upward pressure on interest rates.)
3. **Increase interest rates** → **Increase costs of borrowing** (higher interest rates raise costs of borrowing which subsequently reduces investment expenditure and consequently Aggregate demand falls too)
4. **Decrease aggregate demand** → **Contraction in the economy** (reduced demand leads to economic contraction)

This is why only key D is correct, because as explained above, selling securities do cause interest rates to rise.

27-A: A - A growing economy (increased real GDP) typically leads to increased tax revenues and reduced spending on social safety nets, resulting in a decreased budget deficit. This is because a growing economy usually means higher incomes, profits, and economic activity, generating more tax revenue for the government. Therefore, option A is a likely result of a sustained increase in real GDP.

B - A growing economy typically leads to increased investment in technology, training, and capital, which can improve productivity (output per worker). While it's possible that output per worker might not increase, a decrease is unlikely, especially since the economy is growing.

C - A growing economy can lead to increased imports as incomes rise; this increase in import expenditures may result in a decrease in bop surplus, or may lead to a move towards a deficit.

D- Frictional unemployment arises when workers are temporarily between jobs or searching for new opportunities. A growing economy typically leads to more job opportunities and easier matching of workers with vacancies, reducing frictional unemployment.

28-B:

- A- This option is incorrect because crowding out is a Monetarist concept, not Keynesian. Monetarists believe that government borrowing can lead to crowding out, while Keynesians emphasize the stimulating effect of government spending. The other half, which says that controlling the money supply to control inflation is a monetarist concept, is correct.
- B- Keynesians believe that fiscal policy (government spending and taxation) is more effective for achieving economic growth, while Monetarists believe that wages are flexible and markets can self-correct in the long run. **This is why key B is correct.**
- C- This option is incorrect because it reverses the priorities of both schools. Keynesians typically prioritize reducing unemployment over reducing inflation, while Monetarists prioritize reducing inflation over reducing unemployment.
- D- Keynesians believe that wages are sticky (hard to adjust) downwards, which is why, cyclical unemployment can persist in the long-run. The other half is incorrect as monetarists believe that controlling money supply controls inflation.

29-C:

- The rightward shift in AD (from AD1 to AD2) indicates an increase in aggregate demand, leading to:
 - Higher price level (from P1 to P2)
 - Higher real output (from Y1 to Y2)
- This means the economy is experiencing economic growth, which typically leads to:
 - Increased employment (as businesses expand production and hire more workers)
 - Increased per capita income (as economic growth leads to higher incomes and living standards)
- This is why, option C, becomes correct.

A – As incomes increase, there is likely to be a rise in import spending and when import expenditures rise, current account deficit might increase. Additionally, a rise in demand pull inflation arising from a rise in Aggregate demand may decrease the international competitiveness of export-oriented goods as they now become relatively expensive in the foreign market.

B – Because of a rise in Aggregate demand relative to Aggregate supply, there is likely to be shortages that may translate in a rise in general price level which is termed as demand pull inflation. Additionally, a rise in import expenditures may increase the supply of domestic currency that may result in depreciation of the exchange rate.

D – A rise in Aggregate demand does help to achieve actual economic growth, but rapid increases in economic growth may lead to resources being exploited unsustainably. This means that there will be resource depletion and also environmental degradation.

30- A:

A: Devaluation leads to a decrease in export prices and an increase in import prices . In the long run, when the Marshall Lerner condition holds, export revenues rise and import expenditures fall, resulting in a trade surplus. This, in turn, leads to an increase in Aggregate Demand (AD) relative to Aggregate Supply (AS), causing shortages and demand-pull inflation. Therefore, devaluation would make it difficult to achieve the 3% inflation target, as it would lead to higher inflation due to the demand-pull effect. This is why, key A, is correct.

B: Increasing interest rates would raise costs of borrowing that may reduce Aggregate demand and subsequently demand pull inflation may fall , thereby helping to achieve the target.

C: Increasing the budget surplus would reduce government spending, reducing Aggregate demand and therefore demand pull inflation will fall, making it easier to achieve the target.

D: Keeping wage increases in line with productivity would prevent cost push inflation, and wage price spiral, making it easier to achieve the target.

